A Concrete Lagrangian Derivation of the Schwinger–Dyson Hierarchy

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We give a self-contained operator derivation of the scalar-field Schwinger–Dyson hierarchy, retaining every distributional contact term and every factor of i and $\hbar$. Starting from the free-field mode expansion, we derive the time-ordered propagator, prove that it is the Green function of the Klein–Gordon operator. We then establish the explicit two-point identity $(\Box_x + m^2)G_2(x,y) = \langle\Omega|\mathrm{T}\{[(\Box_x + m^2)\phi(x)]\phi(y)\}|\Omega\rangle - \mathrm{i}\hbar\delta^{(4)}(x-y)$. For an arbitrary product $X_n = \phi(x_1)\cdots\phi(x_n)$, the same argument yields $(\Box_x + m^2)G_{n+1} = \langle\Omega|\mathrm{T}\{\mathcal{L}'_{\rm int}[\phi(x)]X_n\}|\Omega\rangle - \mathrm{i}\hbar\sum_j \delta^{(4)}(x-x_j)G^{(j)}_{n-1}$, where $G^{(j)}_{n-1}$ omits x_j. An independent functional-integral derivation verifies the result. Applications to two-, three-, and four-point functions in cubic and quartic scalar theories provide leading-order checks against Wick contractions. The quantum contact terms arise from differentiating the time-ordering operation, not from modifying the classical operator equation of motion.

I. CONVENTIONS AND SCOPE

We use Minkowski metric $\eta_{\mu\nu} = \mathrm{diag}(+,-,-,-)$, $\Box = \partial_t^2 - \boldsymbol{\nabla}^2$, and $p\cdot x = p^0 t - \boldsymbol{p}\cdot\boldsymbol{x}$. We set $c = 1$ and use natural-unit momentum and mass variables, so plane waves are $\mathrm{e}^{-\mathrm{i}p\cdot x}$ and the Klein–Gordon operator is $\Box + m^2$. We restore $\hbar$ only in quantum normalization and in the path-integral weight. Thus p and m should be understood as the usual natural-unit quantities; a fully dimensionful SI convention would instead place $p_{\rm phys}\cdot x/\hbar$ in the phase and replace the mass term by $(m_{\rm phys}/\hbar)^2$ when $c = 1$. We write the scalar Lagrangian as

$$\mathcal{L} = -\frac{1}{2}\phi(\Box + m^2)\phi + \mathcal{L}_{\rm int}(\phi), \qquad \mathcal{L}'_{\rm int}(\phi) \equiv \frac{\mathrm{d}\mathcal{L}_{\rm int}}{\mathrm{d}\phi}. \quad (1)$$

Up to a surface term the quadratic part is $\frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}m^2\phi^2$. Its Euler–Lagrange equation is therefore

$$(\Box + m^2)\phi - \mathcal{L}'_{\rm int}(\phi) = 0. \quad (2)$$

We define the vacuum-normalized, time-ordered full Green functions by

$$G_N(x_1,\ldots,x_N) \equiv \langle\Omega|\mathrm{T}\{\phi(x_1)\cdots\phi(x_N)\}|\Omega\rangle, \qquad G_0 \equiv 1. \quad (3)$$

Thus our G_2 includes a factor of $\hbar$; it is not the alternative quantity $-\mathrm{i}G_2/\hbar$ also often called the Feynman propagator. This single convention accounts for many apparent sign or normalization disagreements in the literature. The original Green-function program is due to Schwinger [1, 2]; the operator presentation and normalization conventions follow Schwartz [3], while the functional formalism and its modern uses are reviewed in Refs. [4, 5].

The displayed identities are exact identities of the bare, regulated theory. At coincident points, products such as $\phi^3(x)$ are composite operators and require a regulator, counterterms, and possibly operator mixing before a continuum limit is taken. The formal hierarchy survives renormalization, but the Lagrangian parameters and composite insertions must then be understood as renormalized quantities with the appropriate counterterms [4, 5].

II. FREE PROPAGATOR AND TWO-POINT IDENTITY

A. Free field and its time-ordered two-point function

For $E_{\boldsymbol{p}} = \sqrt{\boldsymbol{p}^2 + m^2}$, choose the free-field expansion [3, 6, 7]

$$\phi_0(x) = \sqrt{\hbar}\int \frac{\mathrm{d}^3p}{(2\pi)^3}\frac{1}{\sqrt{2E_{\boldsymbol{p}}}}\left[a_{\boldsymbol{p}}\mathrm{e}^{-\mathrm{i}p\cdot x} + a^\dagger_{\boldsymbol{p}}\mathrm{e}^{\mathrm{i}p\cdot x}\right], \quad (4)$$

where $p^0 = E_{\boldsymbol{p}}$,

$$[a_{\boldsymbol{p}}, a^\dagger_{\boldsymbol{q}}] = (2\pi)^3\delta^{(3)}(\boldsymbol{p}-\boldsymbol{q}), \qquad a_{\boldsymbol{p}}|0\rangle = 0. \quad (5)$$

Standard references usually set $\hbar = 1$. With the oscillator algebra in Eq. (5) left free of $\hbar$, restoring the canonical relation $[\phi_0, \dot\phi_0] = \mathrm{i}\hbar\delta^{(3)}$ requires the overall $\sqrt{\hbar}$ in Eq. (4). An equivalent convention removes that factor and instead places $\hbar$ in the oscillator commutator; the two conventions must not be mixed. Substitution directly gives the equal-time relations

$$[\phi_0(\boldsymbol{x},t), \phi_0(\boldsymbol{y},t)] = 0, \quad (6)$$
$$[\phi_0(\boldsymbol{x},t), \dot\phi_0(\boldsymbol{y},t)] = \mathrm{i}\hbar\delta^{(3)}(\boldsymbol{x}-\boldsymbol{y}). \quad (7)$$

For example, the second line follows because the two surviving oscillator commutators produce

$$[\phi_0(\boldsymbol{x},t), \dot\phi_0(\boldsymbol{y},t)] = \frac{\mathrm{i}\hbar}{2}\int\frac{\mathrm{d}^3p}{(2\pi)^3}\left[\mathrm{e}^{\mathrm{i}\boldsymbol{p}\cdot(\boldsymbol{x}-\boldsymbol{y})} + \mathrm{e}^{-\mathrm{i}\boldsymbol{p}\cdot(\boldsymbol{x}-\boldsymbol{y})}\right] = \mathrm{i}\hbar\delta^{(3)}(\boldsymbol{x}-\boldsymbol{y}). \quad (8)$$

Let $z = x - y$. Expanding the time-ordering symbol,

$$\begin{aligned} D_F(x,y) &\equiv \langle 0|\mathrm{T}\{\phi_0(x)\phi_0(y)\}|0\rangle \\ &= \theta(z^0)\langle 0|\phi_0(x)\phi_0(y)|0\rangle \\ &\quad + \theta(-z^0)\langle 0|\phi_0(y)\phi_0(x)|0\rangle \\ &= \hbar \int \frac{\mathrm{d}^3p}{(2\pi)^3}\frac{\theta(z^0)}{2E_{\boldsymbol{p}}}\mathrm{e}^{-\mathrm{i}E_{\boldsymbol{p}}z^0+\mathrm{i}\boldsymbol{p}\cdot\boldsymbol{z}} \\ &\quad + \hbar \int \frac{\mathrm{d}^3p}{(2\pi)^3}\frac{\theta(-z^0)}{2E_{\boldsymbol{p}}}\mathrm{e}^{\mathrm{i}E_{\boldsymbol{p}}z^0-\mathrm{i}\boldsymbol{p}\cdot\boldsymbol{z}}. \end{aligned} \tag{9}$$

The same distribution has the four-dimensional representation

$$D_F(x,y) = \int \frac{\mathrm{d}^4p}{(2\pi)^4}\frac{\mathrm{i}\hbar\,\mathrm{e}^{-\mathrm{i}p\cdot(x-y)}}{p^2-m^2+\mathrm{i}\epsilon}. \tag{10}$$

To verify Eq. (10), integrate over p^0. The roots of the denominator are

$$p^0 = \pm\sqrt{E_{\boldsymbol{p}}^2 - \mathrm{i}\epsilon} \simeq \pm E_{\boldsymbol{p}} \mp \frac{\mathrm{i}\epsilon}{2E_{\boldsymbol{p}}}. \tag{11}$$

Thus the positive-energy pole lies infinitesimally below the real axis and the negative-energy pole lies infinitesimally above it. For $z^0 > 0$, closing the contour below selects the positive-energy pole; for $z^0 < 0$, closing above selects the negative-energy pole. The two residues reproduce precisely Eq. (9). This is the standard Feynman boundary condition [3, 4, 6, 7].

Now act on Eq. (10) with $\mathcal{K}_x \equiv \Box_x + m^2$. Since

$$(\Box_x + m^2)\mathrm{e}^{-\mathrm{i}p\cdot(x-y)} = -(p^2-m^2)\mathrm{e}^{-\mathrm{i}p\cdot(x-y)}, \tag{12}$$

we obtain, distributionally,

$$\begin{aligned} \mathcal{K}_x D_F(x,y) &= -\mathrm{i}\hbar \int \frac{\mathrm{d}^4p}{(2\pi)^4}\frac{p^2-m^2}{p^2-m^2+\mathrm{i}\epsilon}\mathrm{e}^{-\mathrm{i}p\cdot(x-y)} \\ &= -\mathrm{i}\hbar \int \frac{\mathrm{d}^4p}{(2\pi)^4}\mathrm{e}^{-\mathrm{i}p\cdot(x-y)} = -\mathrm{i}\hbar\delta^{(4)}(x-y). \end{aligned} \tag{13}$$

The ratio in the first line tends to one as a distribution when the Feynman prescription is removed. Thus Eq. (13) directly verifies the contact-term sign.

B. Why derivatives do not simply pass through time ordering

Because $\mathcal{L}_{\mathrm{int}}$ depends on ϕ but not on its derivatives, the exact canonical momentum is $\pi = \dot{\phi}$. The interacting Heisenberg field therefore obeys the same equal-time canonical algebra,

$$[\phi(\boldsymbol{x},t), \phi(\boldsymbol{y},t)] = 0, \tag{14}$$

$$[\phi(\boldsymbol{x},t), \dot{\phi}(\boldsymbol{y},t)] = \mathrm{i}\hbar\delta^{(3)}(\boldsymbol{x}-\boldsymbol{y}). \tag{15}$$

This statement would require modification for derivative interactions, which are outside the scope of the present Lagrangian.

The key identity follows directly from the definition

$$\mathrm{T}\{A(t)B(t')\} = \theta(t-t')A(t)B(t') + \theta(t'-t)B(t')A(t). \tag{16}$$

Using $\partial_t\theta(t-t') = \delta(t-t')$ and $\partial_t\theta(t'-t) = -\delta(t-t')$, differentiation gives

$$\partial_t\mathrm{T}\{A(t)B(t')\} = \mathrm{T}\{\dot{A}(t)B(t')\} + \delta(t-t')[A(t), B(t')]. \tag{17}$$

Set $A = \phi(x)$, $B = \phi(y)$. At the support of $\delta(x^0 - y^0)$, Eq. (15) makes the commutator vanish. Consequently,

$$\partial_{x^0}G_2(x,y) = \langle \mathrm{T}\{\dot{\phi}(x)\phi(y)\}\rangle, \tag{18}$$

which shows explicitly why the first derivative has no contact term.

Differentiate once more and use Eq. (17) with $A = \dot{\phi}$:

$$\partial_{x^0}^2 G_2(x,y) = \langle \mathrm{T}\{\ddot{\phi}(x)\phi(y)\}\rangle + \delta(x^0-y^0)\langle[\dot{\phi}(x), \phi(y)]\rangle. \tag{19}$$

Reversing the canonical commutator in Eq. (15) gives

$$[\dot{\phi}(\boldsymbol{x},t), \phi(\boldsymbol{y},t)] = -\mathrm{i}\hbar\delta^{(3)}(\boldsymbol{x}-\boldsymbol{y}). \tag{20}$$

Therefore

$$\partial_{x^0}^2 G_2(x,y) = \langle \mathrm{T}\{\ddot{\phi}(x)\phi(y)\}\rangle - \mathrm{i}\hbar\delta^{(4)}(x-y), \tag{21}$$

Spatial derivatives do not differentiate a time-ordering step function, so

$$(-\boldsymbol{\nabla}_x^2 + m^2)G_2(x,y) = \langle \mathrm{T}\{(-\boldsymbol{\nabla}_x^2+m^2)\phi(x)\,\phi(y)\}\rangle. \tag{22}$$

Adding Eqs. (21) and (22) yields

$$\boxed{\begin{aligned} (\Box_x + m^2)G_2(x,y) &= \langle \mathrm{T}\{[(\Box_x+m^2)\phi(x)]\phi(y)\}\rangle \\ &\quad - \mathrm{i}\hbar\delta^{(4)}(x-y). \end{aligned}} \tag{23}$$

Equation (23) is the full two-point contact identity. In the abbreviated convention $\langle\cdots\rangle \equiv \langle\Omega|\mathrm{T}\{\cdots\}|\Omega\rangle$, it is exactly

$$\boxed{\begin{aligned} (\Box_x + m^2)\langle\phi(x)\phi(y)\rangle &= \langle[(\Box_x+m^2)\phi(x)]\phi(y)\rangle \\ &\quad - \mathrm{i}\hbar\delta^{(4)}(x-y). \end{aligned}} \tag{24}$$

In the free theory, the operator term on the right vanishes by $(\Box+m^2)\phi_0 = 0$, reducing Eq. (24) to the Fourier-space result (13). Thus the mode-expansion and equal-time-commutator calculations agree exactly.

III. GENERAL CORRELATOR IDENTITY

Let

$$X_n \equiv \phi(x_1)\cdots\phi(x_n), \tag{25}$$

$$X_n^{(j)} \equiv \phi(x_1)\cdots\phi(x_{j-1})\phi(x_{j+1})\cdots\phi(x_n), \tag{26}$$

$$G_{n-1}^{(j)} \equiv G_{n-1}(x_1,\ldots,\widehat{x_j},\ldots,x_n). \tag{27}$$

For bosonic operators, differentiating an $(n+1)$-fold time-ordered product gives the recursive identity

$$\partial_{x^0}\mathrm{T}\{A(x)X_n\} = \mathrm{T}\{\dot{A}(x)X_n\} + \sum_{j=1}^{n}\delta(x^0 - x_j^0)\mathrm{T}\{[A(x),\phi(x_j)]X_n^{(j)}\}. \tag{28}$$

One way to prove this is to divide the time axis into the $(n+1)!$ possible orderings. Within each region the derivative acts only on A. On a boundary $x^0 = x_j^0$, the two adjacent orderings differ by interchanging A and $\phi(x_j)$; their difference is precisely the commutator in Eq. (28). This argument also shows why there is one contact term per spectator field and no extra combinatorial factor.

Taking $A = \phi$, every contact term in the first derivative vanishes:

$$\partial_{x^0}\langle\phi(x)X_n\rangle = \langle\dot{\phi}(x)X_n\rangle, \tag{29}$$

The delta function enforces equal time, at which $[\phi(\boldsymbol{x},t),\phi(\boldsymbol{x}_j,t)] = 0$. Applying Eq. (28) again, now to $A = \dot{\phi}$, gives

$$\begin{aligned}\partial^2_{x^0}\langle\phi(x)X_n\rangle &= \langle\ddot{\phi}(x)X_n\rangle \\ &\quad + \sum_{j=1}^{n}\delta(x^0 - x_j^0)\langle[\dot{\phi}(x),\phi(x_j)]X_n^{(j)}\rangle \\ &= \langle\ddot{\phi}(x)X_n\rangle \\ &\quad - \mathrm{i}\hbar\sum_{j=1}^{n}\delta^{(4)}(x-x_j)\langle X_n^{(j)}\rangle.\end{aligned} \tag{30}$$

In the last line, the spatial delta function in Eq. (20) turns $\boldsymbol{x}$ into $\boldsymbol{x}_j$; the commutator is then a c-number and can be taken outside the remaining time-ordered expectation value. As before, $-\boldsymbol{\nabla}_x^2$ passes directly through the time-ordering operation. Hence

$$\boxed{\begin{aligned}\Box_x G_{n+1}(x,x_1,\ldots,x_n) &= \langle[\Box\phi(x)]X_n\rangle \\ &\quad - \mathrm{i}\hbar\sum_{j=1}^{n}\delta^{(4)}(x-x_j)G_{n-1}^{(j)}.\end{aligned}} \tag{31}$$

Equation (31) is the arbitrary-point contact identity. The hat means that the argument is omitted.

For an explicit three-field check, set $n = 2$ in Eq. (31):

$$\begin{aligned}\Box_x G_3(x,x_1,x_2) &= \langle[\Box\phi(x)]\phi(x_1)\phi(x_2)\rangle \\ &\quad - \mathrm{i}\hbar\delta^{(4)}(x-x_1)G_1(x_2) \\ &\quad - \mathrm{i}\hbar\delta^{(4)}(x-x_2)G_1(x_1).\end{aligned} \tag{32}$$

There are exactly two boundary terms: one when x^0 crosses x_1^0 and one when it crosses x_2^0.

IV. EXACT SCHWINGER–DYSON HIERARCHY

Add $m^2 G_{n+1}$ to both sides of Eq. (31):

$$\begin{aligned}\mathcal{K}_x G_{n+1}(x,x_1,\ldots,x_n) &= \langle[(\Box_x + m^2)\phi(x)]X_n\rangle \\ &\quad - \mathrm{i}\hbar\sum_{j=1}^{n}\delta^{(4)}(x-x_j)G_{n-1}^{(j)}.\end{aligned} \tag{33}$$

The exact Heisenberg operator obeys Eq. (2), namely

$$(\Box_x + m^2)\phi(x) = \mathcal{L}'_{\mathrm{int}}[\phi(x)]. \tag{34}$$

Substitution into only the operator insertion on the right of Eq. (33) gives

$$\boxed{\begin{aligned}\mathcal{K}_x G_{n+1}(x,x_1,\ldots,x_n) &= \langle\mathcal{L}'_{\mathrm{int}}[\phi(x)]X_n\rangle \\ &\quad - \mathrm{i}\hbar\sum_{j=1}^{n}\delta^{(4)}(x-x_j)G_{n-1}^{(j)}.\end{aligned}} \tag{35}$$

Equation (35) is the exact Schwinger–Dyson hierarchy. The first term couples an $(n+1)$-point function to higher correlators; the second line is the quantum contact term. Notice that the operator equation of motion itself has the classical form. The extra term appears because $\Box_x$ acts on the step functions hidden inside T, not because the Heisenberg equation has been altered.

A. Independent functional-integral verification

Equation (35) can be checked without canonical commutators. Start from the normalized Minkowski functional integral and assume a regulator for which the measure is invariant under infinitesimal translations of the field. The integral of a total functional derivative vanishes:

$$0 = \int \mathcal{D}\phi\,\frac{\delta}{\delta\phi(x)}\left[X_n \exp\left(\frac{\mathrm{i}}{\hbar}S[\phi]\right)\right]. \tag{36}$$

The product rule produces

$$0 = \sum_{j=1}^{n}\delta^{(4)}(x-x_j)\langle X_n^{(j)}\rangle + \frac{\mathrm{i}}{\hbar}\langle\frac{\delta S}{\delta\phi(x)}X_n\rangle. \tag{37}$$

From Eq. (1),

$$\frac{\delta S}{\delta\phi(x)} = -(\Box_x + m^2)\phi(x) + \mathcal{L}'_{\mathrm{int}}[\phi(x)]. \tag{38}$$

Multiplying Eq. (37) by $\hbar/\mathrm{i} = -\mathrm{i}\hbar$, using Eq. (38), and rearranging gives exactly Eq. (35). This is the standard Schwinger–Dyson total-derivative identity [1, 2, 4, 5]. Agreement of the operator and functional derivations checks the sign, factor of $\hbar$, number of contact terms, and the argument omitted from each lower correlator.

V. CONCRETE TWO-, THREE-, AND FOUR-POINT APPLICATIONS

A. Two-point function in $\lambda\phi^4$ theory

Take

$$\mathcal{L}_{\rm int} = -\frac{\lambda}{4!}\phi^4, \qquad \mathcal{L}'_{\rm int} = -\frac{\lambda}{3!}\phi^3. \tag{39}$$

With one spectator y, Eq. (35) becomes

$$\boxed{(\Box_x + m^2)G_2(x,y) + \frac{\lambda}{3!}G_4(x,x,x,y) = -\mathrm{i}\hbar\delta^{(4)}(x-y)}. \tag{40}$$

This is an exact equation, but it is not closed: the exact propagator is coupled to a four-field correlator with three coincident insertions.

As a perturbative check, write $G_2 = D_F + G_2^{(1)} + O(\lambda^2)$, where $G_2^{(1)} = O(\lambda)$. Wick's theorem gives

$$G_4^{(0)}(x,x,x,y) = 3D_F(x,x)D_F(x,y). \tag{41}$$

The three arises by choosing which of the three fields at x contracts with the field at y. At order λ, Eq. (40) therefore gives

$$(\Box_x + m^2)G_2^{(1)}(x,y) = -\frac{\lambda}{2}D_F(x,x)D_F(x,y), \tag{42}$$

the position-space tadpole contribution. Its coincident factor $D_F(x,x)$ is ultraviolet divergent in four dimensions, illustrating why the regulated and renormalized qualification in Sec. I is essential.

B. Three-point function in $g\phi^3$ theory

For a genuinely nonzero three-point example, consider the formal perturbative toy model

$$\mathcal{L}_{\rm int} = -\frac{g}{3!}\phi^3, \qquad \mathcal{L}'_{\rm int} = -\frac{g}{2}\phi^2. \tag{43}$$

For real g, its classical potential is unbounded below, so this example is used only to test the hierarchy and its combinatorial factors, not as a nonperturbatively stable theory [5]. Using spectators y, z, the exact equation is

$$\boxed{\begin{aligned}(\Box_x + m^2)G_3(x,y,z) &+ \frac{g}{2}G_4(x,x,y,z)\\ &= -\mathrm{i}\hbar\delta^{(4)}(x-y)G_1(z)\\ &\quad - \mathrm{i}\hbar\delta^{(4)}(x-z)G_1(y).\end{aligned}} \tag{44}$$

If a tadpole renormalization condition enforces $G_1 = 0$, the explicit contact terms vanish, but the equation remains nontrivial because G_3 is coupled to $G_4(x,x,y,z)$.

For a transparent tree-level check, use normal ordering (or cancel tadpoles) and retain connected parts. The first-order connected correlator is

$$G_{3,c}^{(1)}(x,y,z) = -\frac{\mathrm{i}g}{\hbar}\int \mathrm{d}^4u\, D_F(x,u)D_F(y,u)D_F(z,u). \tag{45}$$

The factor 3! from Wick contractions cancels the 3! in the interaction. Applying $\mathcal{K}_x D_F(x,u) = -\mathrm{i}\hbar\delta^{(4)}(x-u)$ gives

$$\mathcal{K}_x G_{3,c}^{(1)}(x,y,z) = -gD_F(x,y)D_F(x,z). \tag{46}$$

On the other hand, after the tadpole pairing is removed, the part of $G_4^{(0)}(x,x,y,z)$ relevant to Eq. (44) has the two contractions $D_F(x,y)D_F(x,z)$, so $(g/2)\times 2D_F(x,y)D_F(x,z) = gD_F(x,y)D_F(x,z)$. This reproduces Eq. (46) exactly.

C. Four-point function in $\lambda\phi^4$ theory

Return to Eq. (39) and take three spectators y, z, w. The exact four-point equation is

$$\boxed{\begin{aligned}(\Box_x + m^2)G_4(x,y,z,w) &+ \frac{\lambda}{3!}G_6(x,x,x,y,z,w)\\ &= -\mathrm{i}\hbar\delta^{(4)}(x-y)G_2(z,w)\\ &\quad - \mathrm{i}\hbar\delta^{(4)}(x-z)G_2(y,w)\\ &\quad - \mathrm{i}\hbar\delta^{(4)}(x-w)G_2(y,z).\end{aligned}} \tag{47}$$

The three contact terms are the three possible ways for the distinguished field at x to meet an external field. The six-point term shows explicitly the infinite-hierarchy structure.

Again the connected tree graph provides a direct check:

$$\begin{aligned}G_{4,c}^{(1)}(x,y,z,w) = -\frac{\mathrm{i}\lambda}{\hbar}\int \mathrm{d}^4u\, D_F(x,u)D_F(y,u)\\ \times D_F(z,u)D_F(w,u).\end{aligned} \tag{48}$$

Acting with $\mathcal{K}_x$ yields

$$\begin{aligned}\mathcal{K}_x G_{4,c}^{(1)}(x,y,z,w) = -\lambda D_F(x,y)D_F(x,z)\\ \times D_F(x,w).\end{aligned} \tag{49}$$

In the free six-point function on the left of Eq. (47), the piece connected to all three remaining external points pairs the three copies of $\phi(x)$ with $\phi(y), \phi(z), \phi(w)$ in $3! = 6$ ways. Thus

$$\frac{\lambda}{3!}G_6^{(0)}(x,x,x,y,z,w) \supset \lambda D_F(x,y)D_F(x,z)D_F(x,w), \tag{50}$$

which, when moved to the right, is exactly Eq. (49). The remaining Wick pairings belong to disconnected or tadpole sectors and are accounted for by the corresponding full correlators and contact terms in Eq. (47).

For completeness, in a $\mathbb{Z}_2$-symmetric vacuum of $\lambda\phi^4$ the odd correlators vanish. Its three-point hierarchy member is consequently

$$(\Box_x + m^2)G_3(x, y, z) + \frac{\lambda}{3!}G_5(x, x, x, y, z) = 0, \qquad (51)$$

because $G_1 = G_3 = G_5 = 0$. This is consistent but contains less physics than the cubic example above.

VI. DISCUSSION AND CHECKS

The derivation can be summarized in three logically separate statements:

1. The Heisenberg field obeys the classical-looking operator equation $(\Box + m^2)\phi = \mathcal{L}'_{\text{int}}$.
2. Two time derivatives of a time-ordered product generate equal-time commutators, and canonical quantization turns them into $-i\hbar\delta^{(4)}$ contact terms.
3. Repeating the calculation for every spectator field produces one contact term per spectator and yields an infinite hierarchy because a nonlinear $\mathcal{L}'_{\text{int}}$ inserts additional fields.

The free Fourier integral, the canonical calculation, the path-integral total-derivative identity, and the explicit tree-level G_3 and G_4 calculations all agree. These are independent checks of the central result, Eq. (35). The hierarchy is exact but is not by itself a closed solution method; practical nonperturbative calculations require a symmetry- preserving truncation and renormalization prescription, as emphasized in the modern Schwinger–Dyson literature [5].

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